

Management of Penetrating Eye Injury with Retained Fish Hook: Case Report

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ABSTRACT

Fishing is commonly practiced around the world as an occupation or for recreation. However, penetrating ocular injuries from fish hooks is uncommon. We report the case of a 14-year-old boy who presented to our facility with 36 hours history of loss of vision on the right eye following injury from fish hook. Examination and investigation revealed a barbed fish hook hanging down from its point of entry into the eye at about 3 o'clock position on the peripheral cornea. There was associated shallow anterior chamber and lens opacity. The cut-it-out technique was employed to successfully remove the retained fish hook. Surgical management of penetrating ocular injury with retained barbed fish hook is challenging as the removal can possibly create more damage to ocular tissues if not extracted correctly. The need for choice of surgical technique for removal of retained fish hook in the anterior segment to be determined on a case-by-case basis taking into cognizance factors such as nature and size of the fish hook and its point of entry into the eyeball is highlighted in this report

Keywords: Penetrating, Eye, Injury, Fishhook

INTRODUCTION

Penetrating eye injury is a form of ocular trauma in which there is a single full thickness laceration of the eyeball usually caused by a sharp object.¹ It is a leading cause of mono-ocular non-congenital visual loss in children.² Up to 40% of penetrating eye injuries are complicated by the presence of retained intraocular foreign body (IOFB).³ The foreign body may be toxic (e.g. iron, copper, vegetable matter) or inert (e.g. glass or plastic). Vision loss may result from the direct mechanical injury or from post-traumatic complications such as endophthalmitis, retinal detachment, metal toxicity and sympathetic ophthalmia.³

Although fishing is highly practiced around the world as an occupation or for recreation, penetrating ocular injury from fish hooks though uncommon, may potentially cause devastating ocular trauma.⁴

Surgical management of penetrating ocular injury with retained barbed fish hook is challenging as the removal can possibly create even more damage to ocular tissues if not extracted correctly. In this report we present the management of a rare case of penetrating eye injury with retained barbed fish hook. The need for choice of surgical technique for removal of retained barbed fish hook in the anterior segment to be determined on case-by-case basis and factors to be considered are also highlighted.

CASE REPORT

A 14-year-old young fisherman presented to our center with approximately 36 hours history of injury to his right eye. He was accidentally struck on his right eye by a fish hook that disengaged from the fish it caught while forcefully pulling on the fishing line. Following the injury, he developed an immediate onset of right ocular pains and loss of vision with the fish hook stuck in the eye. His colleagues' attempts to remove the fish hook stuck in his eye were unsuccessful. On clinical examination, his uncorrected presenting visual acuity was hand movement

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on the right eye and 6/5 on the left. A fish hook was seen hanging down from its point of entry into the eyeball at about 3 o'clock position on the peripheral cornea approximately 1.5mm from the limbus (Figure 1). Part of the hook was seen running obliquely in the anterior chamber from 3 o'clock position of entry at the limbus toward 6 o'clock position but the tip was not clearly visible. There was associated mild conjunctival hyperemia, cornea edema, shallow anterior chamber (AC), hairline hyphaema, lens opacity and iris tear involving the inferior-nasal quadrant. The pupil was irregular, mid dilated and unreactive to light. Fundoscopy revealed an absent red reflex. The left eye was normal.

Ocular plain radiograph (X-ray) revealed the intraocular extent of the fish hook and single barbed nature of the hook (Figure 2) Hematological and biochemical tests including packed cell volume and random blood sugar were conducted. The results were within normal limits.



Figure 1 (A) Patient at presentation (B) At first day post removal of retained fish hook.

The patient was admitted and worked up for emergency removal of the retained fish hook under general anesthesia. Pre-operatively, he was commenced on prophylactic broad spectrum parenteral antibiotics and analgesics and received anti-tetanus prophylaxis. At surgery, after routine

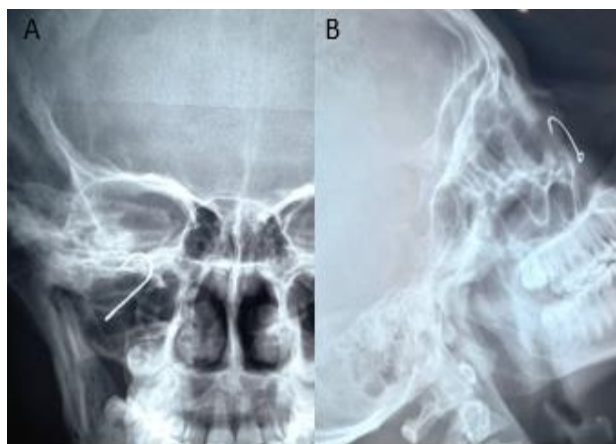


Figure 2. Plain X-ray. (A) anterior-posterior and (B) lateral views showing the barbed nature of the fish hook.



Figure 3. Picture of the barbed fish hook after removal from patient's eye.

cleaning of periocular skin and conjunctival cul-de-sac with 10% and 5% povidone iodine respectively, sterile linen was used to drape the operating field. The shallow AC was then deepened with ophthalmic viscoelastic device (OVD) introduced through a side port created on clear cornea about 2mm from 9 o'clock position on the limbus. The entry wound of the fish hook on the cornea was then widened using a side port knife in a controlled manner. The fish hook was subsequently pulled out in backward fashion through the widened entry wound (figure 3). The widened entry wound was then closed using a 9/0 nylon suture in interrupted fashion. The AC was then washed out and reformed using balance salt solution.

Sub-conjunctival dexamethasone and gentamycin was administered at the end of the surgical procedure.

Post operatively topical dexamethasone and ciprofloxacin were given four times daily and continued for 6 weeks. At 6 weeks follow up, the AC was well formed and the traumatic cataract has become more prominent. Cataract surgery was planned for a later date.

DISCUSSION

Penetrating fish hook eye injury is rare but can potentially cause devastating ocular damage and complications. The severity of damage caused by a penetrating fish hook eye injury depend on several factors including the nature of the fish hook, its velocity and location in the eyeball. Ocular complications reported due to fish hook injury include traumatic cataract, cornea scarring, hyphema, choroidal and vitreous hemorrhages, retinal tears and detachment and endophthalmitis.⁴ Our patient was found to have iris tear, traumatic cataract and hyphaema at presentation. It is likely that the unsuccessful attempt at removal of the hook before presentation may have caused more injury to the eye.

Removal of retained barbed fish hook from delicate ocular tissues can be quite challenging. Simply pulling out the fish hook from the eye can cause more damage than the initial injury due to the barb engaging and traumatizing surrounding tissues.^{5,6}

Several methods of fish hook removal from ocular and non-ocular tissues described in literature^{4,5} include: (a) Back-out or retrograde technique. Here, the fish hook is backed out of the entry wound. This method is employed for barbless fish hooks as the presence of barb will cause increased tissue injury while backing out. (b) Advance-and-cut technique. The hook is pushed forward until the tip and barb are outside the eye. Subsequently, wire cutters are used to cut the hook between the barb and the bend. The shank is then pulled out of the eye

through the entry wound using the back out technique described previously. (c) Snatch technique. This is a modified form of back out technique employed for removal of fish hook from non-ocular tissues. In this method, downward pressure on the hook shank and rapid back out extraction are used to reduce pain during removal. (d) Needle-cover technique. This method has been reported when the fish-hook penetrates the retina. A large bore needle is passed into the eye through the entry wound. The barb is covered with the lumen of the needle to prevent it from entangling in the tissues and both are then withdrawn together. (e) Cut-it-out technique. In this method, the entrance wound is enlarged by a scalpel blade. Then, the blade is slid along the hook until it reaches the barb. Subsequently, the hook is backed out of the eye similar to the back-out technique.

From the forgone, choice technique for removal of retained fish hook from the eyeball majorly depends on its nature (barbed or barbless) and its location in the eyeball (anterior or posterior segment involving the retina). Generally, simple back out technique is used for the removal of retained penetrating non barbed fish hook ocular injury. However, the ideal technique for removal of a barbed fish hook (especially in the anterior segment as in this case) is debatable.

The advance and cut technique have been touted as the ideal approach for barbed fish hook⁶ with several cases of good outcomes reported.^{7,8} Its advantages include surgically controlled second wound, no enlargement of primary wound and minimal traumatic manipulation.⁶ It is particularly considered most ideal for central cornea wounds whereby removal of the barbed fish hook through a surgically controlled peripheral cornea wound allows for repair of less astigmatic inducing wound and minimization of additional iatrogenic central cornea trauma.⁹ Disadvantages of the advance and cut technique include potentially increased risk of

infection from creation of iatrogenic wound with fish hook that is often contaminated and not being suitable for small hooks because they require significant manipulation to push through compared to a hook with large diameter bend. Also, this method requires the use of wire cutters in the operating room which ophthalmologist rarely have access to, or have experience using.⁹

On the other hand, the cut it out technique has the advantage of being suitable for small and large fishhooks and does not require use of wire cutters that are not readily available in the ophthalmic operating room. However, it is not suitable when the entry point of the hook is on the central cornea due to potential for higher refractive errors associated with enlarging and suturing of entry point wound on the central cornea.⁹

In this patient the cut it out technique was employed to successfully remove the retained fish hook in the anterior segment. We considered this choice most appropriate for this case because of the peripheral nature of the entry point on the cornea. Besides we do not have wire cutters or experience using it. Malitz.¹⁰ was the first to publish use of this technique successfully in an ocular setting in 1993, and several other cases have been reported with similarly positive outcomes.^{4,6}

Conclusion

Management of penetrating eye injury with retained barbed fish hook is potentially challenging. Choice technique for removal of barbed fish hook retained in the anterior segment should be determined on a case-by-case basis. Factors such as nature and size of the barbed fish hook, its point of entry (whether through central or peripheral cornea), the tools available to and experience of the ophthalmologist should be considered before deciding on the best approach for removal of the fish hook.

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